P.Raja Raghava Reddy

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Education

2023 - 2027 Bachelor's Degree at SRM Institute of Science and Technology (GPA: 8.95/10.0)

2021 - 2023 Class 12 - Shree Swaminarayan Gurukul Vidyalaya (74)

2020 - 2021 Class 10 - Shree Swaminarayan Gurukul International School (74)

Experience

AgroScope - Intern Feb 2026 - May 2026

Projects

AI-Based Temple Parking Management

Tech Stack: Python, YOLOv8, OpenCV, FastAPI, Firebase Realtime Database, Leaflet.js

- Developed a real-time CCTV-based parking management system using YOLOv8 and OpenCV to detect vehicles and monitor parking occupancy across multiple camera feeds.
- Implemented CLAHE, polygon-based ROI masking, and multithreaded frame processing to handle varying lighting conditions, filter irrelevant traffic, and stabilize vehicle detection.
- Built a FastAPI backend for multi-camera video streaming and integrated Firebase Realtime Database and Leaflet.js to display live parking availability on an interactive web dashboard

Intelligent Defect Detection

Tech Stack: Python, OpenCV, NumPy

- Developed a system that processes industrial images to automatically identify and locate surface defects such as cracks, scratches, and corrosion.
- Used 8-connected component analysis to group connected defect pixels into individual regions and extract important properties such as area, shape, and size.
- Applied image preprocessing, feature analysis, and defect classification to separate actual defects from noise and generate clear inspection results.

Resume-Job Description Matcher

Tech Stack: Python, scikit-learn, PyMuPDF, Gradio, MLflow, pytest

- Built an ML tool that scores how well a resume matches a job description, extracting text from PDF resumes with PyMuPDF and computing TF-IDF n-gram cosine similarity.
- Implemented boundary-aware regex skill matching against a curated skills catalog to avoid false positives (e.g., Java vs. JavaScript) and to list matched and missing skills.
- Blended text similarity and skill coverage into a weighted 0-100 score, tuned with MLflow experiment tracking and validated with pytest, served through a Gradio interface with actionable resume feedback.

Skills

Languages and Web: JavaScript, Python

Backend and Databases: FastAPI, MySQL

ML and Tools: scikit-learn, OpenCV, YOLOv8, NumPy, MLflow, Gradio, pytest

Certificates

Introduction to Machine Learning NPTEL

Database Management System NPTEL

Natural Language Processing NPTEL

Languages

Hindi Telugu English